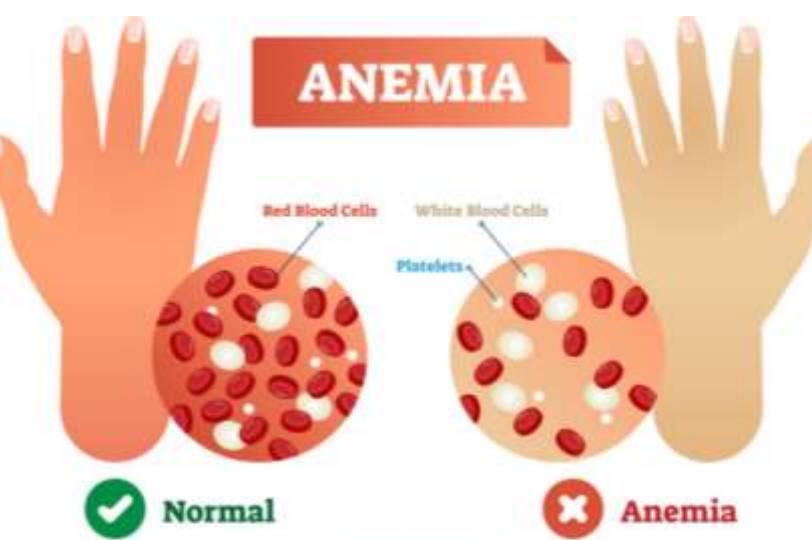
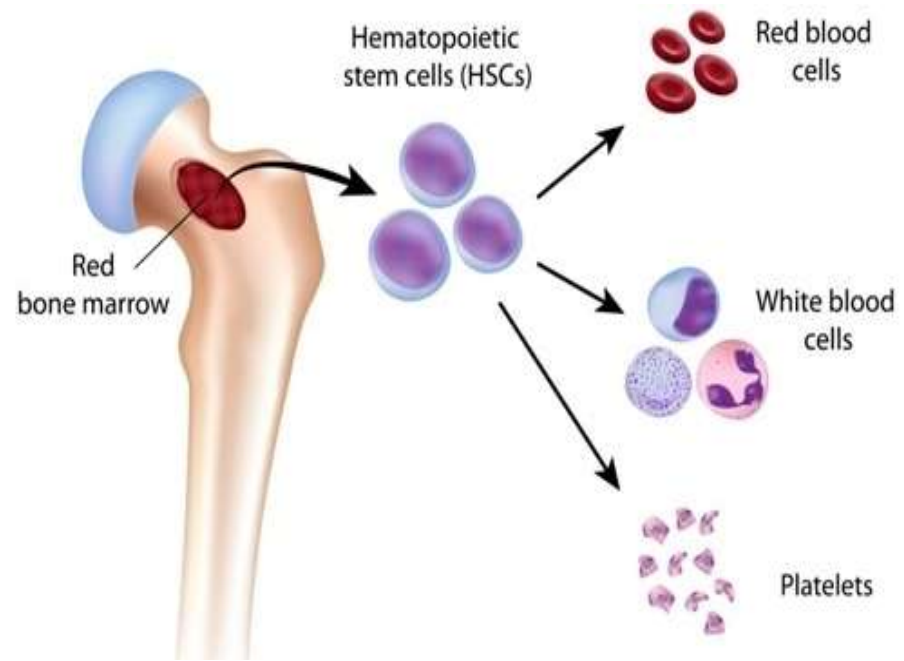




Dr. HANAA ELSAYED

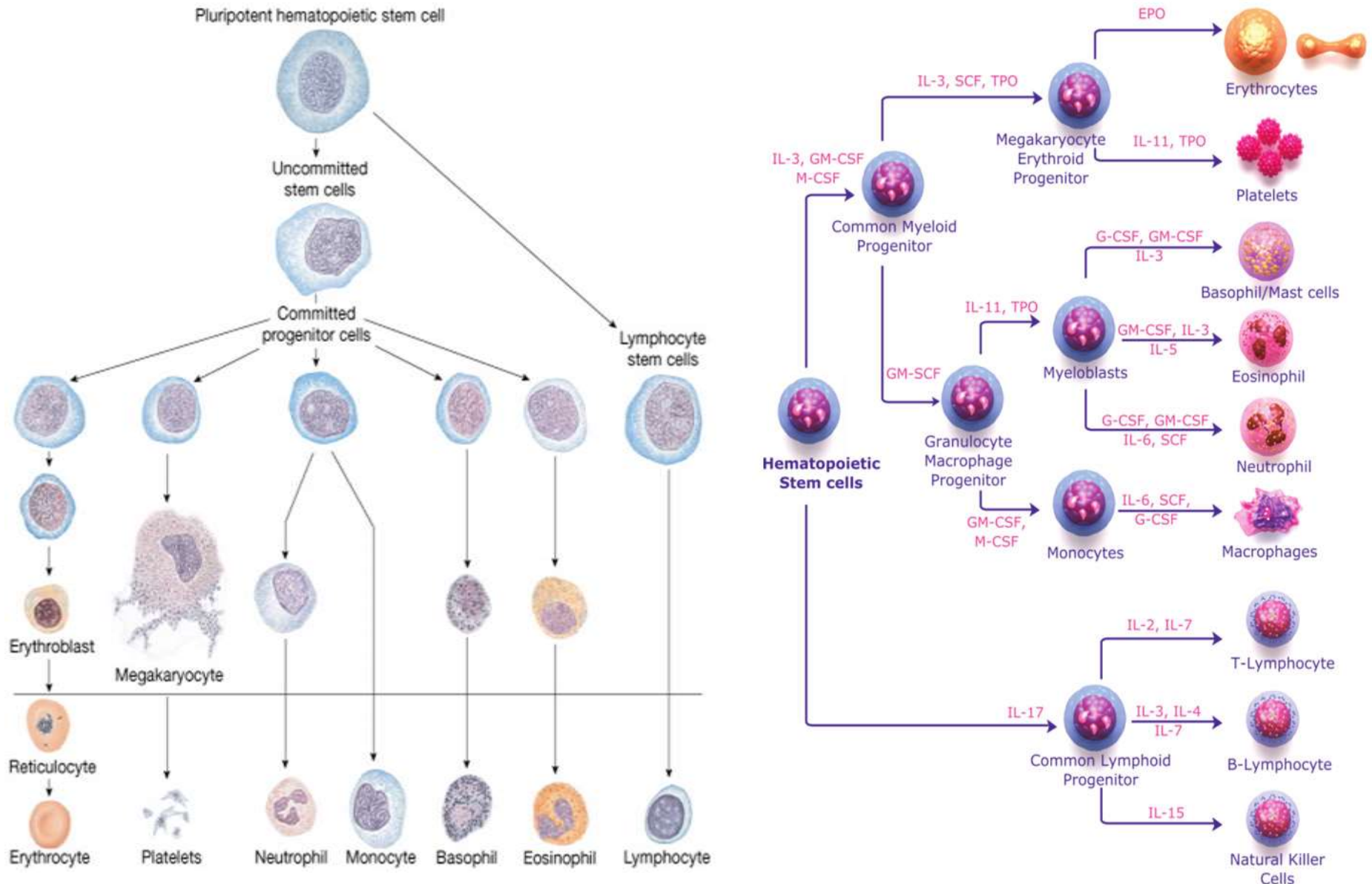
ANEMIA



ILOs

- **By the end of this lecture, students should be able to:**
 - **Define anemia**
 - **Classify anemia**
 - **How to interpret with each type of anemia**
 - **Dental aspect of different type of anemia**
 - **Construct management of anemia**

Hematopoiesis (Blood Cell Formation)



Anemia

- It is a common clinical problem in medicine.
- It is not a disease but may be a feature of many diseases.
- **Definition:** decrease Hb level below the reference level for the age and sex of the individual.
- **Causes:**
 - Red blood cell (RBC) **production** is inadequate or
 - RBC lifespan (normally 120 days) is shortened through loss from the circulation or **destruction**.

	Male	Female
Hb (g/dL)	13.5–17.5	11.5–16
PCV (haematocrit; L/L)	0.4–0.54	0.37–0.47
RCC ($10^{12}/L$)	4.5–6.0	3.9–5.0
MCV (fL)	80–96	
MCH (pg)	27–32	
MCHC (g/dL)	32–36	
RDW (%)	11–15	
WBC ($10^9/L$)	4.0–11.0	
Platelets ($10^9/L$)	150–400	
ESR (mm/h)	< 20	
Reticulocytes	0.5–2.5% ($50–100 \times 10^9/L$)	

ESR, erythrocyte sedimentation rate; Hb, haemoglobin; MCH, mean corpuscular haemoglobin; MCHC, mean corpuscular haemoglobin concentration; MCV, mean corpuscular volume of red cells; PCV, packed cell volume; RCC, red cell count; RDW, red blood cell distribution width; WBC, white blood count.

Etiological Classification

- **Decreased red cell production**
 - **Stem cell failure:** as Aplastic anemia
 - **Progenitor erythroid cell failure:** as Anemia of chronic disease
 - **Precursor cell failure:** as Iron deficiency and megaloblastic anemia.
- **Increased red cell destruction or loss**
 - **Hereditary:**
 - Hereditary spherocytosis, thalassemia, sickle cell anemia and glucose 6 phosphate dehydrogenase (G6PD) deficiency.
 - **Acquired:**
 - Immune hemolytic anemia, Hypersplenism, Microangiopathic hemolytic anemia e.g. disseminated intravascular coagulopathy (DIC) and Acute blood loss.

Morphological Classification according to the MCV (N: 80 -100 fl)

- **Microcytic anemia [< 78 fl]**
 - Iron deficiency anemia
 - Thalassemia
 - Anemia of chronic disease
 - Sideroblastic anemia
- **Normocytic anemia [79-98 fl]**
 - Hemolytic anemia
 - Acute post hemorrhagic anemia
 - Stem cell failure
 - Anemia of chronic disease or organ failure
- **Macrocytic anemia [>99 fl]**
 - Megaloblastic anemia as vitamin B12 or folic acid deficiency
 - Non-megaloblastic
 - Myelodysplastic syndrome (ineffective hematopoiesis)
 - Hypothyroidism
 - Liver disease.
 - Alcoholism
 - Drugs: methotrexate.

Clinical Picture of Anemia

- **General:**

- Fatigue (tiredness), dizziness, headache and syncope.

- **Cardiorespiratory:**

- Dyspnea, chest pain, palpitation.
- Congestive heart failure.
- Murmurs, tachycardia, bounding pulse.

- **Cutaneous:**

- Pallor.
- Brittle nails.
- Koilonychias (spoon-shaped nails – iron deficiency).



- **Oral:**

- Sore mouth or oral ulceration.
- Angular stomatitis.
- Glossitis.
- Burning mouth symptoms.



Management

- **Complete blood count (CBC), Erythrocyte sedimentation rate (ESR):**
 - **Hb level, RBCs and blood indices.**
- **Reticulocyte count.**
- **Blood film:** Abnormal RBC
 - **Sickle cells in sickle cell disease.**
 - **Pencil cells in iron-deficiency anemia.**
- **Hemoglobin electrophoresis:** hemoglobinopathy screening.
- **Serum vitamin B12, folic acid, iron and ferritin levels**
- **Endoscopy:** gastrointestinal blood loss.
- **Bone marrow biopsy.**

- **Treatment of anemia:**

- According to type, severity and onset of anemia:

- If acute **and severe** anemia determine treated with **transfusion therapy** with **packed RBCs**.

- Hct should increase 3–4% [Hb by 1 g/dL] with **each unit of packed RBCs**.

- It Prevent worsening of ischemic symptoms.

- **Risk** of infection, circulatory overload and allergic reactions.

- **Chronic anemia** (vitamin B12 deficiency), even when severe, **may not require transfusion therapy, however** and specific therapy (Vit. B12) is instituted.

- **Hematinic such as iron or folic acid.**

- **Erythropoietin is mainly used in:**

- Treatment of anemia of chronic renal failure or

- Cytotoxic therapy.

- **Anemia should be corrected pre-operatively:**
 - **Whole blood transfusion** given to a young and otherwise fit patient.
 - **Packed red cells** used in older patient or those with congestive heart failure (CHF).
 - To avoid the risk of fluid overload.
 - **Treatment:** **diuretic** to **reduces** the risk of CHF.

- **Local anesthesia** is usually satisfactory for pain control.
- **GA** is given with **full oxygenation**.
 - Elective operations under **GA** should not usually be carried out when the Hb is **less than 10 g/dL** (male).
 - The patient should be stabilized at least **24 hours preoperatively**.

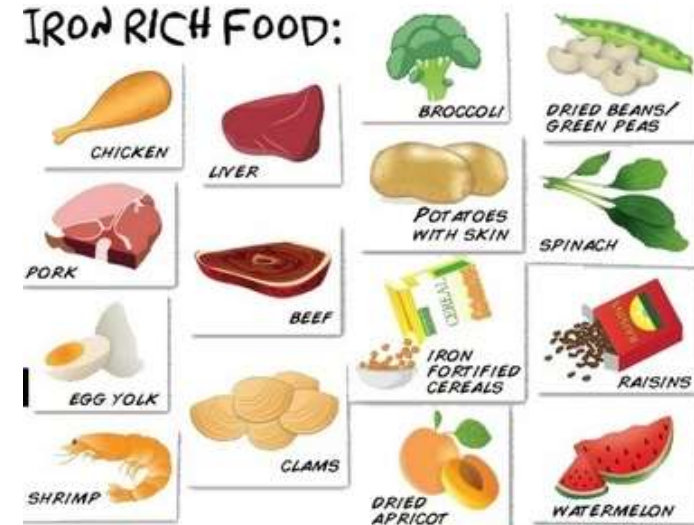
Iron deficiency anemia (IDA)

- Is the most cause of anemia, affect more than 3.5 billion people.

- **Etiology:**

- **Increased blood loss:**
 - Usually from gastrointestinal (peptic ulcer).
 - Excessive menstrual blood loss
- **Decreased iron intake:**
 - Seen in infants, old individuals.
- **Decreased iron absorption:**
 - Phytates, oxalates and phosphates present in plant foods, and tannin in tea, can inhibit iron absorption.
- **Increased iron requirements** as in females in childbearing period, children and adolescents

IRON RICH FOOD:



Clinical picture

- Clinical picture of anemia.
- Features of iron deficiency:
 - Impairment of growth and intellectual development in **children.**
 - **Blue discoloration of sclera** (due to thinness and transparency of the collagen fibers of the sclera that allows visualization of the underlying uvea)
 - Koilonychias, angular stomatitis, and **Plummer-Vinson syndrome.**
- Clinical picture of the cause.

Differential Diagnosis of Microcytic Anemia

Tests	Iron Deficiency	Inflammation	Thalassemia	Sideroblastic Anemia
Smear	Micro/hypo	Normal micro/hypo	Micro/hypo with targeting	Variable
SI	<30	<50	Normal to high	Normal to high
TIBC	>360	<300	Normal	Normal
Percent saturation	<10	10–20	30–80	30–80
Ferritin (µg/L)	<15	30–200	50–300	50–300
Hemoglobin pattern on electrophoresis	Normal	Normal	Abnormal with β thalassemia; can be normal with α thalassemia	Normal

Abbreviations: SI, serum iron; TIBC, total iron-binding capacity.

Treatment

- Treat cause.
- **Oral iron** may need to be given for 3 months or more
 - **Ferrous sulfate** 200 mg three times daily orally.
 - **Ferrous gluconate** 250 mg/day can be given if ferrous sulfate is not tolerated.
 - **Side effects:** Nausea or constipation, black stool.
- **Parenteral iron:** given intramuscularly.
 - Used if the patient cannot take oral iron or when inflammatory bowel disease is present (aggravated by oral iron)
 - **Side effect:** skin reaction, arrhythmias
- **Anemia of chronic disease:**
 - Treat underlying disease.
 - In uremia: recombinant human erythropoietin, 50–150 U/kg three times a week .

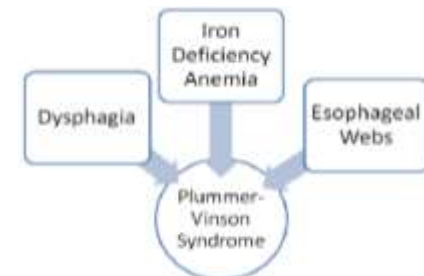
Dental Aspects of Iron Deficiency Anemia

- Local anesthesia is satisfactory for pain control.
- Conscious sedation can be given if full oxygenation is possible.
- **Plummer–Vinson syndrome, also known as Patterson–Brown–Kelly syndrome: Uncommon.**



– Clinically:

- Atrophic glossitis or angular cheilitis
- Dysphagia due to pharyngoesophageal ulcerations and esophageal webs
- Iron deficiency anemia.
- Risk of carcinoma in the post-cricoid region or in the mouth.



- **Atrophic glossitis, soreness or burning sensation of the tongue with depapillation with dysfunction of taste sensation.**
- **Various type of candidiasis.**
- **Angular cheilitis or stomatitis (frequently caused by infection, mainly *Candida albicans*).**
- **Recurrent aphthous ulcers (RAU).**
- **Other miscellaneous oral manifestations include hyperpigmentation and oral lichen planus (OLP).**
- **Staining of the teeth especially in children:**
 - **Prevented by using sodium iron edetate as the iron source, as it is also sugar free and more palatable than ferrous sulfate.**
- **Some iron preparations can cause tooth erosion.**



Megaloblastic anemia

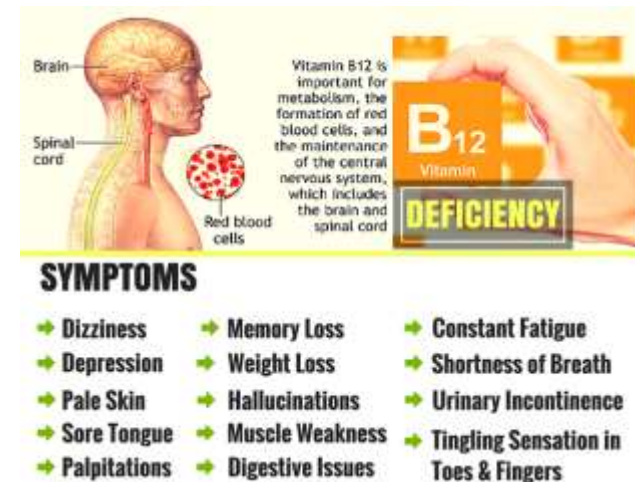
- **Vitamin B12 and folic acid:** needed to synthesize DNA/RNA and needed to build new cells (blood, skin and mucosal cells).
- Vitamin B12 is bound to **gastric intrinsic factor** (secreted by parietal cells) and **absorbed in the terminal ileum**.
- It is **stored in the liver** (Because a large amount of vitamin B12 is stored in the liver, it takes 5-10 years for clinical problems to manifest following decreased intake or absorption of vitamin B12).

- Vitamin B12 is contained in animal foods, and the daily intake is approximately 3-30 μg . The daily required amount is approximately 1-3 μg .
- The recommended dietary allowance of folic acid for adults is 240 μg a day.
- There are virtually **no body stores of folic acid**.
- Both Vit. B12 and folate deficiency cause **megaloblastic changes in bone marrow, defective DNA synthesis** and impaired production of blood cells and other cells.

	Vitamin B12	Folic Acid
Sources	Meat, especially liver.	
Absorption	Terminal part of intestine	Proximal small intestine
Causes of deficiency	- Defect in intrinsic factor (pernicious anemia, atrophic gastritis, or gastrectomy). - Occasionally ileal disease or resection. - Vegetarianism. - Drugs: Colchicine, Neomycin, Nitrous oxide, Omeprazole. 	- Poor intake: Poverty, Old age, Alcoholism - Malabsorption: Coeliac, Crohn disease. - Increased demands: Infancy, Pregnancy, chronic disease, Chronic dialysis. - Drugs: Alcohol, Azathioprine, Methotrexate, Sulfasalazine 

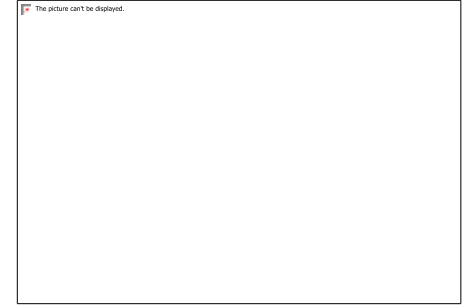
Clinical Pictures

- Clinical picture of anemia
- **Gastrointestinal:** Soreness of the tongue, aphthous ulcers, weight loss, diarrhea or constipation.
- **In severe condition:**
 - Neurological symptoms (vitamin B12 deficiency only):
 - **Sub acute combined degeneration of the spinal cord,** paraesthesia of the extremities, deep sensory loss and difficult walking.
 - Memory problems, irritability, ataxia, decreased taste, decrease level of consciousness, depression, anxiety, guilt and psychosis.



- **Diagnosis:**

- **CBC: anemia**
- **Peripheral blood smear: shows a macrocytic anemia, hypersegmented granulocytes with a mild leukopenia and thrombocytopenia.**
- **Mean cell volume (MCV) and mean cell hemoglobin (MCH) are increased**
- **Vitamin B12 (200-900 pg/ml), Folic acid (2.5-50 ng/ml)**
- **Indirect bilirubin level and serum lactate dehydrogenase (LDH) concentration may be elevated (pernicious anemia).**
- **Autoantibodies against gastric parietal cells and/or intrinsic factor.**



Treatment

- Most patients have malabsorption and will require parenteral or high-dose oral replacement.
- **Vitamin B12 deficiency:** vitamin B12 100 µg IM for 7 d, then 100–1000 µg IM/ month or with 2 mg oral crystalline vitamin B12 / day
- **Pernicious anemia** is treated for life with intramuscular hydroxycobalamin 1 mg.
- **Folate deficiency:** folic acid 1 mg PO (5 mg for pts with malabsorption).

Dental aspects

- Local anesthesia is satisfactory.
- **Conscious sedation** can be given if the hemoglobin level is only moderately depressed and supplemental oxygen can be given.
- **Nitrous oxide is (theoretically) contraindicated** (Nitrous oxide can also interfere with vitamin B12 metabolism).
- **GA should be postponed** until low hemoglobin has been remedied.
- In early B12 deficiency, the **tongue is sore or burning tongue**.
- **Candidosis**: can be aggravated or precipitated by anemia and may be the presenting feature.
- **Angular stomatitis is uncommon**.
- **Aphthous stomatitis is occasionally the presenting feature and pernicious anemia should be considered when ulceration starts in middle age or later.**

Hemolytic Anemia

- **Causes:**

- **INHERITED:**

- **Inherited abnormalities of Hb formation (hemoglobinopathies):**

- **Thalassaemia, sickle cell disease.**

- **Membrane abnormalities (rare):**

- **Hereditary spherocytosis:** autosomal dominant (Spherocytes are more rigid and less deformable)
 - **Hereditary elliptocytosis:** autosomal dominant (Clinically, it is a similar condition to HS but milder).

- **Metabolic defects:**

- **Glucose-6-phosphate dehydrogenase [G6PD] deficiency:**

- **Common condition and affects millions of people throughout the world, particularly Africa, Middle East.**

- **Risk factors:**

- » **Drugs Aspirin, Sulfonamides, Quinolones.**

- » **Ingestion of fava beans.**

- **Acquired:**

- **Autoimmune.**

- **Non-immune:**

- **Acquired membrane defects as Paroxysmal nocturnal haemoglobinuria (PNH).**
 - **Microangiopathic hemolytic anemia (MAHA).**
 - **Malignant hypertension.**
 - **Thrombotic thrombocytopenic purpura (TTP), Hemolytic-uremic syndrome (HUS).**

- **Others:**

- **Infections (e.g., malaria).**

- **Characteristic features:**

- Jaundice due to **excessive bilirubin**.

- Spleen may enlarge.

- **Increase reticulocyte count.**

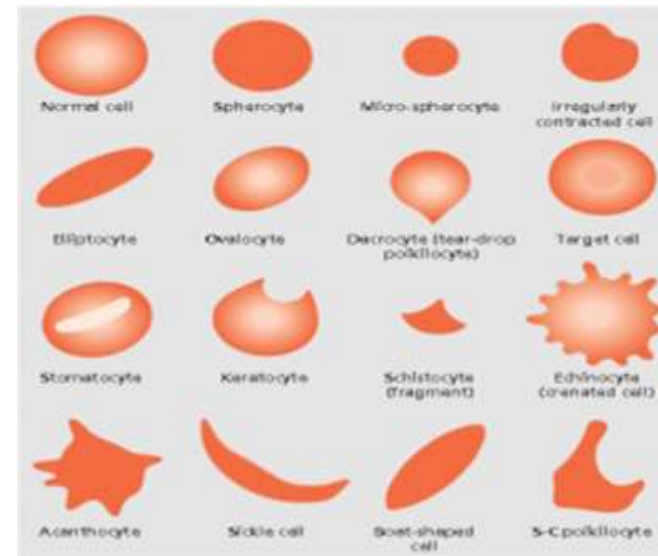
- Elevated **unconjugated bilirubin**, uric acid and LDH, low or absent haptoglobin (Intravascular hemolysis).

- **Coombs' test:** immuno-hemolytic anemia.

- **Osmotic fragility test** (hereditary spherocytosis)

- **Hemoglobin electrophoresis** (sickle cell anemia, thalassemia)

- G6PD assay.

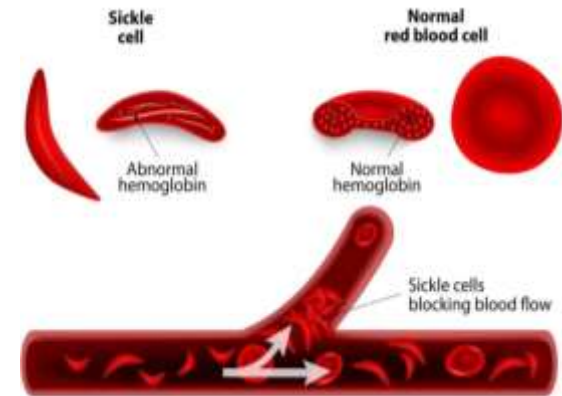


Sickle cell anemia (SCA)

- It includes sickle cell anemia or disease, and sickle cell trait.
- **Autosomal recessive.**
- **Caused by** replacement of glutamic acid by valine the beta-chain of globin, thus resulting in hemoglobin S.
 - **Sickle cell anemia or disease (HbSS):**
 - Homozygous and is usually a serious disease with widespread complications.
 - Frequently apparent in about the third month of life.
 - **Sickle cell trait (HbSA):**
 - Heterozygous and at least ten times as common.
 - Sickle cell trait is frequently asymptomatic.
- **Anemia and chronic hemolysis.**
- The major manifestations relate to vasoocclusion from sickled RBCs.

Normal hemoglobins:

- Fetal HbF: 2Alpha, 2 Gamma
- Adult HbA: 2Alpha, 2Beta
- HbA2 2Alpha, 2 Delta



- **Hematological crises:** vasoocclusive, hemolytic, aplastic or sequestration.
 - **Precipitated factors:**
 - Infection, dehydration, hypoxia, acidosis or cold.
- Infarcts mainly in the spleen (autosplenectomy), **bones and joints** (especially hands and feet), brain, kidneys, lungs, eye and skin.
 - **Pulmonary infarcts:**
 - **Chest pain** and eventually lead to pulmonary hypertension and right sided cardiac failure.
 - **Kidneys:**
 - Causing hematuria and nephrotic syndrome.
 - Ocular defects or cerebrovascular stroke.

Orofacial manifestations of SCA

- Mucosal pallor, yellow tissue coloration, delayed tooth eruption and tooth fractures.
- **Periodontitis** that is unusual in children.
- Disorders of **enamel and dentine mineralization**.
- Enlarged maxilla with excessive over jet, and overbite.



Fractured incisors

• Pallor of the oral mucosa.

• Slow tooth development and eruption.

• Hypoplasia of enamel secondary to anemia.

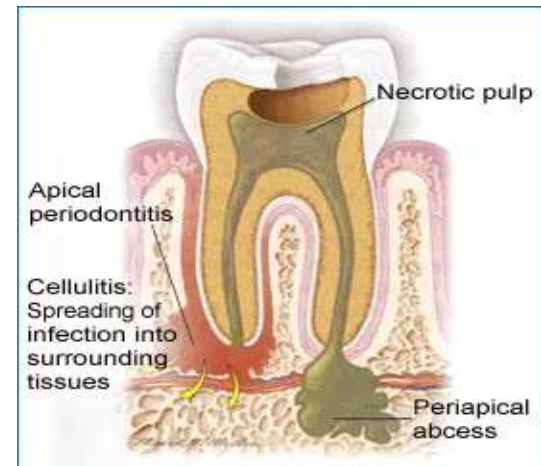
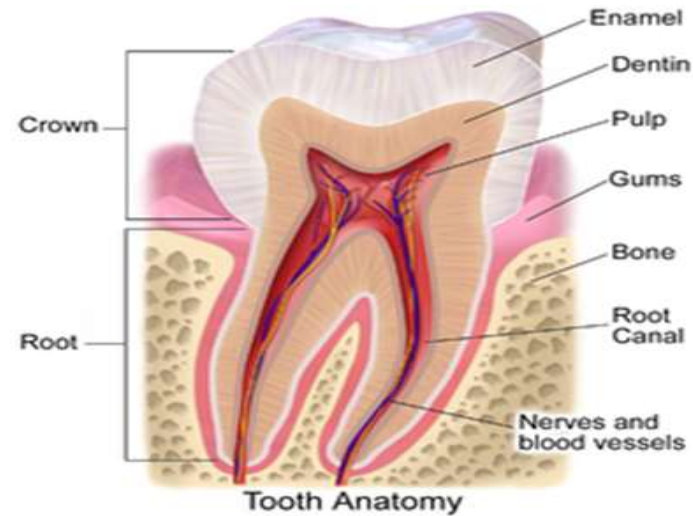


INCREASED OVERJET

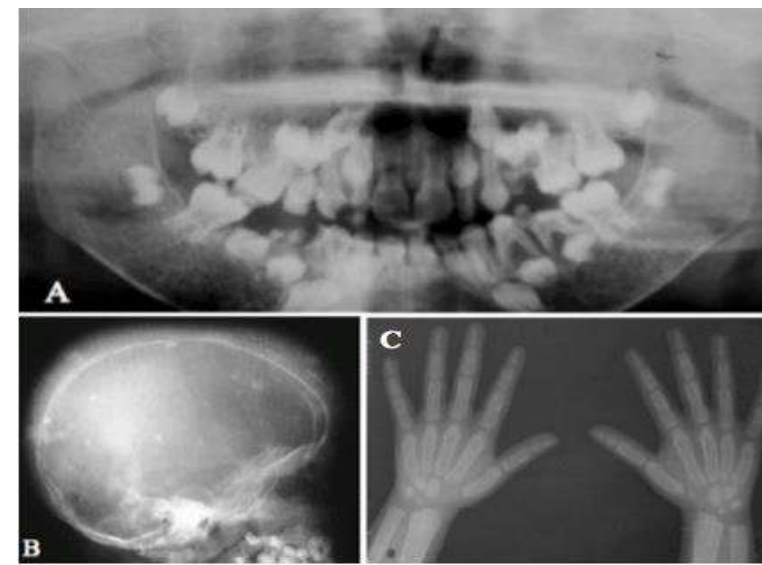
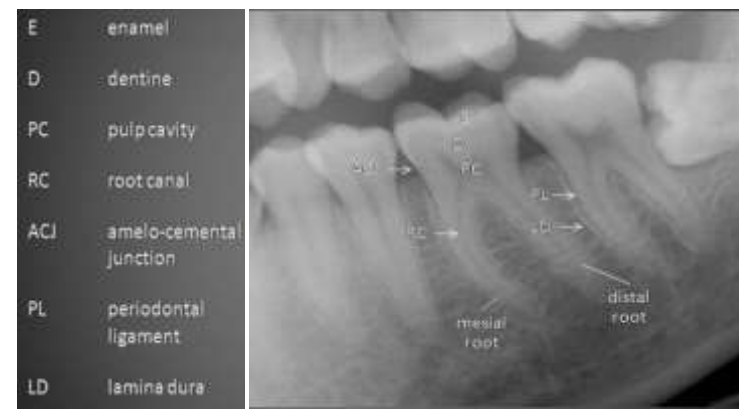


INCREASED OVERBITE

- **Analgesic abuse** and addiction during crises.
- **Painful infarcts in the jaws** (mistaken for toothache) or osteomyelitis.
- **Pulpal symptoms are common in absence of any dental disease and sometimes **pulpal necrosis** has resulted.**



- **Radiographic findings:** ‘stepladder’ trabeculae pattern.
- Bone infarction (dense radio opacities) may be seen in the jaws and/or skull.
- **Skull:** thickened but osteoporotic with a hair-on-end pattern to the trabeculae.
- The lamina dura is distinct and dense, and the permanent teeth may be hypo mineralized.



Management

- **Hb level:** below 9 g/dL.
- **Blood film:** target cells, and sometimes sickled erythrocytes
- **Hb electrophoresis:**
 - HbS and up to 15% HbF, but no HbA.
 - 40% HbS and up to 60% HbA in sickle cell trait.
- **Treatment:**
 - Treat risk factors as dehydration, infection, hypoxia.
 - Folic acid.
 - Blood transfusions.
 - Analgesics, such as diamorphine or pethidine.
 - **Hydroxyurea:** 10–30 mg/kg / day PO raises the levels of HbF and prevent sickling.
 - **Iron-chelating agents:** desferrioxamine.
- **Prognosis:** infections and thrombosis are the main causes of death.

Dental aspects of SCA

- **LA is the preferred of pain control.**
- **Prilocaine** (local anesthetic) is best avoided, may in overdose, cause methemoglobinaemia.
- **Aspirin is best avoided** as, in large doses, it may cause acidosis and precipitate a crisis.
 - **Paracetamol and codeine are effective.**
 - **Acidosis and hypotension must also be avoided.**
- **Conscious sedation** with relative analgesia can be used safely. At least 30% O2 is needed and, **provided that there is no respiratory depression** or obstruction, normal procedures can be used.
 - **Benzodiazepines are best avoided.**
 - **Drugs that can cause respiratory depression**, including sedative agents, should not be given.

- **Elective surgery** should be carried out in hospital and during hemolysis is minimal.
 - **Anemia should be corrected** before GA and the hemoglobin brought up to at least 10 g/dl.
 - **As a consequence of multiple transfusions**, some patients have become infected with blood-borne viruses.
- **Exchange transfusion** is required for major surgery in selected patients.
- **Packed red cell transfusion.**
- **Prophylactic antimicrobials** (penicillin V or clindamycin) should be given for surgical procedures (one hour before).
 - **Infections must be treated vigorously**, since the patient may be immunocompromised if the spleen is non-functional.

Thalassemia

- Autosomal dominant disorder.
- Affect people living in the Mediterranean basin but now worldwide.
- Many types exist the most common is B thalassemia
- B thalassemia there is either **absence of B chains** production or **impairment of B chain production**.

Oro-facial features of thalassemia

- **Enlargement of the upper jaw** (chipmunk face)
- **Migration and spacing of upper anterior teeth.**
- Varying degrees of **malocclusion** (overbite, open bite).
- Teeth may be **discolored**, with short crowns and roots.
- Higher rate of dental decay.
- **Pale gums and mucosa** / lining of the mouth (due to anemia).
- **Sore or burning tongue** (due to folate deficiency).
- **Painful swelling of salivary glands** and **dry mouth** and reduced salivary protection (reduced IgA)



Migration and spacing of the upper anterior teeth



Chipmunk facies along with malaligned teeth in upper and lower

j: Beth Thalassemia Major – bone changes



Dental Aspects of Thalassemia

- LA is safe. CS may be given with oxygen levels not less than 30%.
- GA induction may be complicated by enlargement of the maxilla, which may cause difficulties in intubation.
- GA is best avoided in chronic severe anemia and cardiomyopathy (due to effects of iron deposition).
- Prophylactic antibiotics to reduce the risk of infection following any procedures (most notably dental extractions or scaling) especially in patient has had a splenectomy.
- Recurrent exchange transfusions: increase risk of blood-borne virus infections as HBV, HIV.

Aplastic Anemia



- Aplastic anemia is a serious and often fatal hematologic disorder characterized by peripheral **pancytopenia** with **hypoplastic or non functioning bone marrow**.
- It is a rare (incidence of 2-6/1,000,000) disease causing leukopenia, thrombocytopenia and refractory anemia.
- It is more common in Asian countries than in the United States and Europe with about 6000–7000 new diagnosis reported annually worldwide.
- It can appear at any age but is most commonly diagnosed in children aged 2–5 years, young adults between 20 and 25 years and adults aged 55–60 years.

Causes and risk factors

- Genetic.
- Acquired:
 - Chemicals such as benzene.
 - **Drugs:** Allopurinol, Anticonvulsants, Cytotoxic agents, NSAIDs, Penicillamine, Sulphonamides.
 - **Infections:** hepatitis viruses.
 - Irradiation.
 - Unknown (more than 50 %)

- **Clinical pictures:**

- Clinical picture of anemia, bleeding due to thrombocytopenia and recurrent infections due to neutropenia



- **Management:**

- The diagnosis ‘aplastic anemia’ is confirmed by hypocellularity of the bone marrow. Treat cause.
- Isolation and antibiotics to control infection.
- Bone marrow transplantation (BMT); haemopoietic stem cell transplantation, HSCT).
- Corticosteroid.
- **Blood transfusion** – but this carries risks from iron overload or infection.

- **Oral manifestations of aplastic anemia are common:**
 - Petechial hemorrhages, Pallor, ulceration, hemorrhagic tendencies, and susceptibility to infections.
 - Gingivitis and periodontitis
- Oral lichenoid lesions, or a Sjogren-like syndrome, may develop if there is graft-versus-host disease (GVHD) following marrow transplantation to relieve the Anemia.
- **Gingival swelling** may develop if ciclosporin is used.



Dental Aspects

- Dental management of patients of aplastic anemia requires interdisciplinary care with the consultation of the treating dentist with hematologist.
- It is advisable to perform dental treatment on the day of platelet transfusion.
- Local anesthesia is satisfactory.
- **Conscious sedation and GA** should be avoided because of the anemia.
- Reduce the risk of uncontrolled bleeding during major dental treatments, the patients should take antifibrinolytics.
- Oral hygiene instruction, dental prophylaxis and systemic aminocaproic acid.
- **Dental treatment should be postponed** until the patient's white blood cell count rises to a normal level.
- Dentists should consider prescribing **antibacterial mouthwash** and oral antibiotics before dental procedures.

- Which of the following deficiency is a cause of microcytic anemia?
- A. Folic acid.
- B. Iron.
- C. Vitamin B12.
- D. Vitamin C.

- During tooth extraction of a patient with sickle cell anemia he develops a vaso-occlusive crisis. What is the immediate management?
- A. Oral aspirin.
- B. O2 therapy.
- C. IV antibiotics.
- D. IV heparin

- In which of the following types of anemia the use of nitrous oxide is contraindicated?
- A. B12 deficiency.
- B. Iron deficiency.
- C. Aplastic.
- D. Sick cell.

- Radiographic findings of stepladder trabeculae pattern and calcified pulp canal are found in which type of anemia?
- A. Aplastic.
- B. Thalassemia.
- C. Sick cell.
- D. Megaloblastic.

Thank you!!!

